

CO 5420 - Project Plan

Fabric Defect Detection and Localisation Using Deep Neural Networks

1 Group Details

Field	Details
Module Code	CO 5420
Project Title	Fabric Defect Detection and Localisation Using Deep Neural Networks
Dataset	AITEX Fabric Image Database
Dataset Source	https://www.kaggle.com/datasets/nexuswho/aitex-fabric-image-database
Submission Type	Project Draft Plan
Group Members	F.R.Sujeewan(E/22/382) R.Dakshayani(E/22/060) N.Sathurjika(E/22/261) K.Nithilaa(E/22/260) R.Shathursima(E/22/330)
Semester	4
Submission Date	19/06/2026

2 Motivation / Problem Definition

Fabric quality control is a critical stage in the textile manufacturing industry. Even minor defects—such as holes, loose threads, stains, and weaving irregularities—can significantly reduce product value and lead to costly wastage. Traditional manual inspection methods are slow, inconsistent, and prone to human error, particularly when handling high production volumes under variable lighting and surface conditions.

This project addresses the challenge of automatically detecting and localising defects in high-resolution fabric images using a combination of classical image processing and modern deep learning techniques. The overall goal is to develop a robust automated inspection pipeline that:

- Accurately classifies fabric images as defective or defect-free (binary classification);
- Segments and localises the precise spatial extent of any detected defect at the pixel

level;

- Explores unsupervised anomaly detection approaches that do not require labelled defective training examples.

Automating fabric defect detection has direct industrial impact: faster throughput, reduced material waste, higher consistency, and scalable quality assurance. This project also serves as a rich applied computer vision case study, spanning classical image analysis, supervised deep learning, transfer learning, and unsupervised anomaly detection within a constrained real-world dataset.

2.1 Research Questions

1. Can classical image processing techniques (texture features, edge detection) provide a viable baseline for fabric defect classification?
2. How much does a CNN improve performance over the classical baseline, and which training strategies (data augmentation, transfer learning, patch-based training) are most effective under limited data conditions?
3. Can an autoencoder trained exclusively on defect-free images detect defects via reconstruction error, without any defective samples during training?
4. How accurately can a U-Net segmentation model localise defects using the provided binary masks, and how does its localisation quality compare with the autoencoder's anomaly maps?

3 Dataset Description

This project uses the **AITEX Fabric Image Database**, a publicly available dataset specifically designed for fabric defect detection and localisation research. The dataset is accessible via AITEX and Kaggle.

3.1 Dataset Statistics

Property	Value	Notes
Total Images	247	High-resolution grayscale images
Defect-Free Images	141	Used for normal training (autoencoder)
Defective Images	106	Cover 9 defect categories
Image Dimensions	4096 × 256 px	Very wide panoramic strips
Fabric Types	7	Various weave patterns
Defect Categories	9	Holes, stains, threads, picks, etc.
Mask Images	1 per defective image	Binary ground truth for segmentation
Availability	Free / Open	AITEX official site and Kaggle

3.2 Dataset Characteristics and Challenges

The AITEX dataset presents several notable characteristics and challenges that directly inform our methodological choices:

- **Small dataset size:** With only 247 images total, this is a low-data regime requiring careful regularisation, data augmentation, and transfer learning strategies.
- **High image resolution (4096 × 256 px):** The panoramic strip format means images must be divided into patches for CNN training; direct resizing would lose critical texture detail.
- **Class imbalance:** 141 defect-free vs. 106 defective images. Class weighting or oversampling may be needed to prevent classifier bias.
- **Intra-class variation:** Seven different fabric types with distinct textures increase classification difficulty and require models to generalise across fabric styles.
- **Pixel-accurate ground truth:** Binary mask images provided for all defective samples enable supervised segmentation training and rigorous evaluation via Intersection over Union (IoU).
- **Rare defect occurrence:** Defect regions typically occupy a small spatial fraction of each image, creating pixel-level class imbalance during segmentation training.

3.3 Proposed Data Splits

Splits are applied stratified across fabric types and defect categories:

Split	Defect-Free	Defective	Primary Use
Training (70%)	≈ 99	≈ 74	Model training for all tasks
Validation (15%)	≈ 21	≈ 16	Hyperparameter tuning, early stopping
Test (15%)	≈ 21	≈ 16	Final evaluation only (held out)

4 Proposed Contribution (Division of Work)

All members will participate in final integration, report writing, and presentation preparation. The table below outlines individual contributions.

Member Name	Planned Contribution / Task	Approx. Timeline
K.Nithilaa	Task 1: Classical baseline (LBP/GLCM + SVM and HOG + Random Forest); data preprocessing, train/validation/test splits, dataset analysis, exploratory visualisations, feature extraction, and baseline performance evaluation using accuracy, confusion matrix, precision, recall, and F1-score.	Weeks 1–2
N.Sathurjika	Task 2: CNN classifier implementation (ResNet-50 and/or EfficientNet-B0 transfer learning); patch extraction, data augmentation pipeline, hyperparameter tuning, training strategy development, and evaluation using Accuracy, F1-score, and AUC-ROC.	Weeks 2–4
R.Dakshayani	Task 3: Convolutional/Variational Autoencoder development; anomaly-scoring pipeline, reconstruction error analysis, threshold optimisation, anomaly-map generation, image-level and pixel-level evaluation, and comparison against supervised methods.	Weeks 3–5
R.Shathursima	Task 4 (Part A): U-Net segmentation architecture development; mask preprocessing, patch generation, implementation of standard U-Net and pretrained U-Net variants, encoder-decoder design, skip-connection configuration, and preparation of the segmentation training pipeline.	Weeks 4–5

F.R.Sujeevan	Task 4 (Part B): U-Net model training and optimisation; implementation of Dice Loss and Binary Cross-Entropy loss, augmentation strategies, hyperparameter tuning, IoU and Dice coefficient evaluation, localisation analysis, and comparison of U-Net segmentation maps with autoencoder anomaly maps.	Weeks 5–6
All Members	Report writing, codebase integration, result collation, experimental analysis, presentation slides preparation, documentation, testing, and peer review of contributions.	Weeks 6–7

5 Generative AI Usage Statement

The following generative AI tools may be used during the project development process.

AI Tool Planned to Use	Purpose / How It Will Be Used
ChatGPT	Assistance with project planning, algorithm selection, implementation guidance for machine learning and deep learning models, code suggestions, troubleshooting, report drafting, documentation improvements, and presentation preparation.
Claude	Assistance with literature review, technical explanations, methodology discussions, code debugging, documentation improvements, and structured report writing support.

Declaration: By submitting this Draft Plan, the group confirms that the content reflects the group's own understanding and decisions, and that any generative AI assistance has been disclosed above in accordance with the module's plagiarism policy.

6 References

- [1] A Public Fabric Database for Defect Detection Methods and Results: https://www.researchgate.net/publication/334068330_A_Public_Fabric_Database_for_Defect_Detection_Methods_and_Results
- [2] AITEX Fabric Image Database. Available at: <https://www.kaggle.com/datasets/nexuswho/aitex-fabric-image-database>
- [3] Research on Textile Defect Detection Algorithm for Deep Learning: <https://www.sciencedirect.com/science/article/abs/pii/S0010482522006783>
- [4] Fabric Defect Detection Using Deep Convolutional Neural Network: https://www.researchgate.net/publication/355091253_Fabric_Defect_Detection_Using_Deep_Convolutional_Neural_Network